

Face Sketch to Photo Synthesis using Pix2Pix GAN on the CUFS Dataset

INTRODUCTION

Face sketch to photo synthesis is an image-to-image translation problem where the goal is to generate a realistic face photograph from a hand-drawn sketch. This task is useful in applications such as law enforcement and identity recognition, where sketches are often available instead of real photos.

In this project, I focus on translating face sketches into face photographs using a Pix2Pix conditional Generative Adversarial Network (GAN). Pix2Pix is a supervised deep learning model that learns a mapping between paired images. The experiments are conducted using the CUHK Face Sketch Database (CUFS), which contains corresponding sketch and photo image pairs.

The objective of this part of the project is to train a Pix2Pix model to learn sketch-to-photo mapping and to analyze how the generated images improve during training.

DATASET DESCRIPTION

The dataset used in this project is the CUHK Face Sketch Database (CUFS), obtained from Kaggle. The dataset consists of paired images, where each sketch image has a corresponding real face photograph of the same person.

In total, 188 image pairs were used in this project. The images are frontal face images captured under controlled conditions. The dataset was manually organized into two folders: one containing sketch images and the other containing face photographs.

The dataset was split into training and validation sets using an 80%–20% ratio. The paired nature of the dataset makes it suitable for supervised image-to-image translation using the Pix2Pix framework.

```
ROOT = Path("CUHK Face Sketch Database (CUFS)")

PHOTOS_DIR = ROOT / "photos"
SKETCHES_DIR = ROOT / "sketches"

print("Photos folder exists:", PHOTOS_DIR.exists())
print("Sketches folder exists:", SKETCHES_DIR.exists())
print("Photo jpg count:", len(list(PHOTOS_DIR.glob("*.jpg"))))
print("Sketch jpg count:", len(list(SKETCHES_DIR.glob("*.jpg"))))
```

```
Photos folder exists: True
Sketches folder exists: True
Photo jpg count: 188
Sketch jpg count: 188
```

METHODOLOGY – PIX2PIX GAN

Pix2Pix is a conditional Generative Adversarial Network (cGAN) designed for supervised image-to-image translation tasks. It consists of two main components: a generator and a discriminator. The generator learns to transform an input image (sketch) into a target image (photo), while the discriminator learns to distinguish between real photos and generated images.

In this project, the generator is implemented using a U-Net architecture, which includes an encoder-decoder structure with skip connections. These skip connections help preserve low-level spatial information from the input sketch. The discriminator follows a PatchGAN architecture, which classifies image patches instead of the entire image, encouraging the generation of realistic local details.

The generator is trained using a combination of adversarial loss and L1 loss. The adversarial loss helps generate realistic images, while the L1 loss ensures that the generated image is structurally similar to the ground truth photo.

```

bce = tf.keras.losses.BinaryCrossentropy(from_logits=True)
LAMBDA = 100

g_opt = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)
d_opt = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)

def gen_loss(d_fake, gen_out, target):
    gan = bce(tf.ones_like(d_fake), d_fake)
    l1 = tf.reduce_mean(tf.abs(target - gen_out))
    return gan + LAMBDA * l1

def disc_loss(d_real, d_fake):
    real = bce(tf.ones_like(d_real), d_real)
    fake = bce(tf.zeros_like(d_fake), d_fake)
    return real + fake

```

IMPLEMENTATION DETAILS

The Pix2Pix model was implemented in TensorFlow using a Jupyter Notebook environment. All images were resized to 256×256 pixels and normalized to the range [-1, 1]. The Adam optimizer was used for both the generator and the discriminator with a learning rate of $2e-4$ and $\beta_1 = 0.5$.

```

g_opt = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)
d_opt = tf.keras.optimizers.Adam(2e-4, beta_1=0.5)

```

The model was trained for 100 epochs. During training, one validation image pair was used to generate a preview at the end of each epoch. These preview images were saved in order to visually track how the generated results evolved over time.

```

EPOCHS = 100

```

The training and evaluation process was implemented entirely in a Jupyter Notebook (.ipynb) environment, following the same workflow used during the course lectures and labs.

EXPERIMENTAL RESULTS

During training, the model gradually improved its ability to generate facial photos from sketches. In the early epochs, the generated images appeared blurry and lacked clear facial details. As training progressed, facial structure such as eyes, nose, and hair became more recognizable.

To monitor performance, generated images were saved at each epoch using a fixed validation sample. Additionally, the generator loss was tracked to identify the best-performing model. The best results were achieved at epoch 69, which produced the lowest generator loss.

The best-performing generator model and its corresponding output image were saved separately to ensure that the final results reflect the highest visual quality achieved during training.

```
# -----  
best_g = float("inf")  
best_epoch = -1  
BEST_MODEL_PATH = ROOT / "pix2pix_generator_best.keras"
```

```
if gl_value < best_g:  
    best_g = gl_value  
    best_epoch = e + 1  
  
# Save model  
G.save(str(BEST_MODEL_PATH))
```

Sketch



Real Photo



Generated



Image 1 – Epoch 1

Sketch



Real Photo



Generated



Image 10 – Epoch 10



Image 50 – Epoch 50



Image 69 – Epoch 69 (Best)



Image 100 – Epoch 100 (Final Epoch)

To monitor performance, generated images were saved at each epoch using a fixed validation sample. In addition to visual inspection, the generator loss was observed during training to identify the best-performing model. The best results were achieved at Epoch 69, which produced the lowest generator loss and visually showed the most realistic facial structure.

DISCUSSION

Although the Pix2Pix model successfully learns the general structure of faces, the generated images still lack sharp details and appear slightly blurry. This behavior is expected due to the limited size of the dataset and the complexity of the sketch-to-photo translation task.

GAN training is known to be unstable, and generator loss values do not always decrease smoothly. For this reason, visual inspection of the generated images was used in addition to loss values to select the best model. The results show that Pix2Pix can learn meaningful mappings (face shape, hair region, and skin tone) even with a relatively small dataset.

CONCLUSION

In this project, a Pix2Pix conditional GAN was applied to the task of face sketch to photo synthesis using the CUFS dataset. The model was trained for 100 epochs and showed gradual qualitative improvement over time.

The final results demonstrate that Pix2Pix is effective for supervised image-to-image translation when paired data is available. While the generated images are not perfectly realistic, they preserve important facial structures and show clear improvement during training. Future work could include training on larger datasets or using more advanced GAN architectures to improve image quality.